

FINAL PRESENTATION

What NVIDIA's job postings reveal

1,745 postings · 12 weekly observations · 13 May to 2 September 2026

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Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

THE HEADLINE

Three findings

- **NVIDIA is not expanding. It is turning its open roles over faster.** ~416 roles open at any time, +4.6% in four months, while arrivals and closures both rose ~25%.
- **NVIDIA hires the people who build the machine, not the people who use it.** Hardware, systems and silicon: 885 of 1,793 skill mentions. ML frameworks: 14.
- **The rotation is toward the field.** Product and go-to-market 13.7% → 18.8%; research 16.1% → 11.3%. Solutions Architecture is the largest single skill.

And one finding about the method

Two of the ten tasks produced results I am recommending the team does NOT publish yet. Section 8 explains why that is the most useful thing in this deck.

TASK 1

The legal finding that shaped every task after it

NVIDIA's careers host **permits** crawling in robots.txt. Section 3.2 of NVIDIA's Terms of Service **prohibits** robots, scrapers, crawlers and data-mining tools.

Where the two disagree, the Terms of Service govern

robots.txt is an instruction to crawlers. The Terms of Service are the contractual position. Checking robots.txt alone would have led me to collect from a source I am not permitted to use.

Consequence: no first-party collection, so no description text - which constrained Tasks 3, 4, 6 and 8. Every limitation in this deck traces back to this slide.

TASK 2

Both approved sources failed on contact with the data

Source	Expected	Actual	Verdict
Kaggle LinkedIn postings	24 months of history	4-week snapshot, 11 NVIDIA postings	Rejected
HF data_jobs	cross-company basis	271 NVIDIA rows, data roles only, no text	Rejected
Aramente/eu-tech-jobs	found during collection	1,745 postings, 12 weekly observations, CC BY-4.0	Used

The Kaggle token was never the obstacle: the dataset simply has 11 NVIDIA rows. I also discarded my first pipeline - the source's own diff files report 8,797 "new" events for 2,029 postings and never once mark one removed.

TASKS 3 & 4

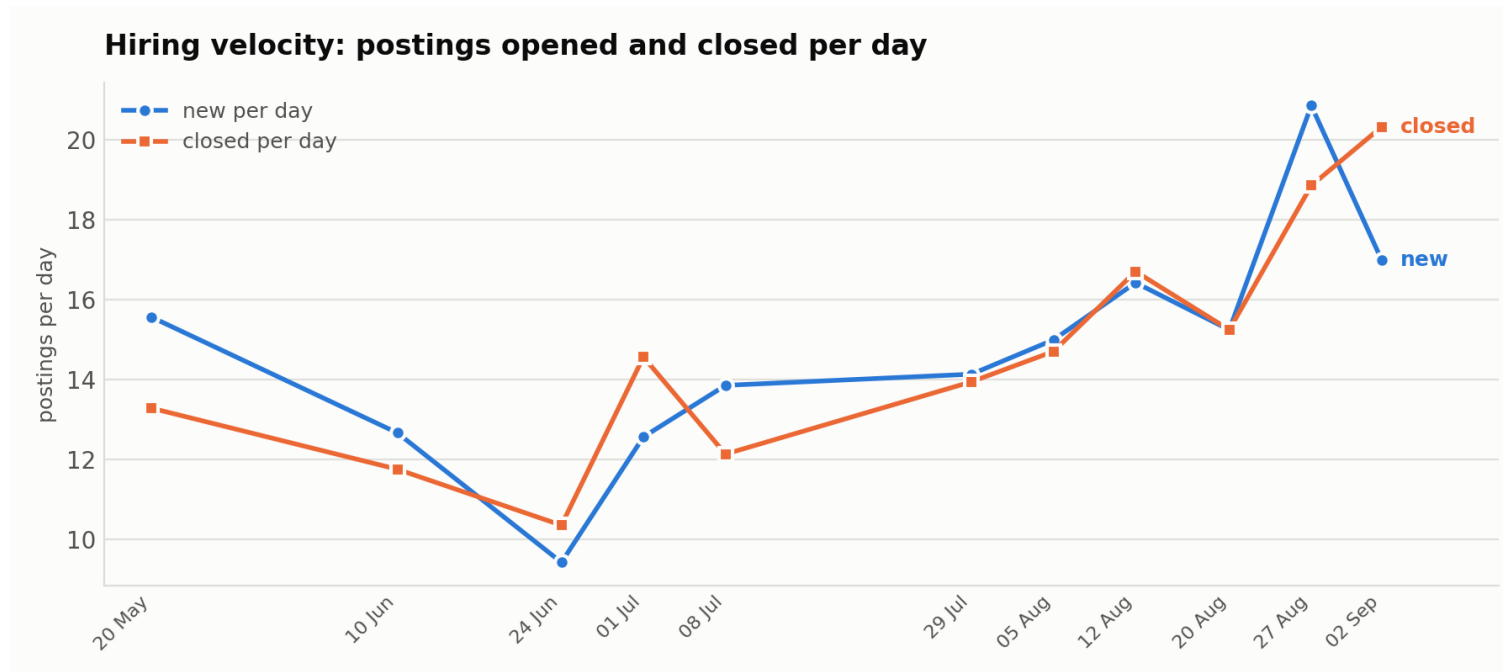
Titles, not descriptions - and what that cost

	NVIDIA	Google	Meta	Microsoft
Skills per posting, titles	1.03	0.34	0.17	0.08
Skills per posting, full text	no text	2.43	4.13	6.84

- Standard preprocessing would destroy a six-word title, so the pipeline **parses title structure** instead of reducing text. Seniority coverage 662 → 1,229; role function 878 → 1,743.
- The shared taxonomy covered only **21.2%** of NVIDIA postings - it lists tools that belong in a description. I added 34 skills and a silicon_design category: coverage → **72.0%**.
- Hand-checking 20 postings caught a false positive and a miss worth **192 postings** (titles say "solutions architect", the taxonomy said "solutions architecture").

TASK 5

Flat stock, accelerating churn

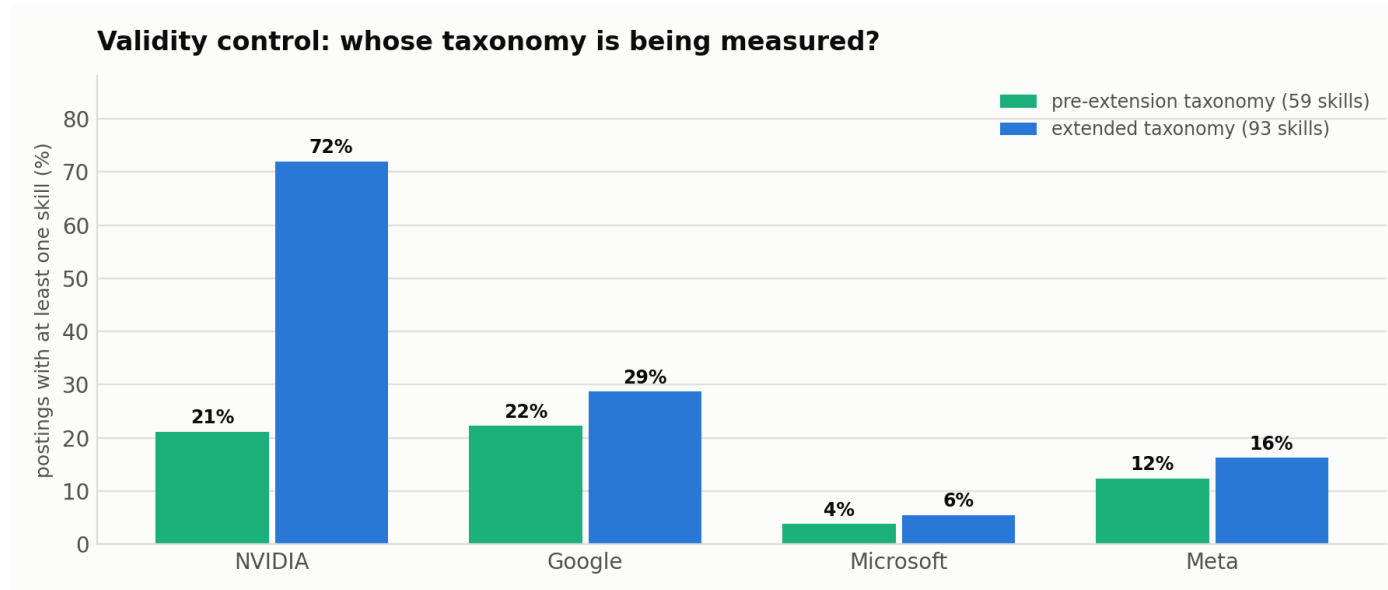


New and closed track within **3.3 postings per day at every observation** while both rise ~25%. Replacement, not expansion.

Flows are reported per day: the observation gaps run 6-21 days, and raw counts turned the two 21-day gaps into fake spikes of 266 and 297 arrivals - which normalise to 12.7 and 14.1.

TASK 6

The result I am asking the team not to publish



NVIDIA's entire apparent advantage is my own taxonomy work

Under the taxonomy the team agreed, NVIDIA scores 21.2% and Google 22.3% - NVIDIA is behind. Under my extension NVIDIA gains 50.8 points and the others gain 1.7 to 6.4. The 34 skills I added were built by reading NVIDIA titles, so they find NVIDIA vocabulary. Until every member does the same for their company, a four-way comparison measures the taxonomy, not the companies.

TASK 7

What the backtest caught

Series	Best model	MAE	Naive MAE	12-week forecast
Open roles	Simple exponential smoothing	5.91	9.00	416 (377-455)
New per day	Mean of last 3	1.99	3.02	17.7 (7.6-27.8)

Task 5's upward trend does not forecast

Task 5 measured arrivals rising at +0.60 per observation. In backtesting, the two models that extrapolate a trend - Drift and Holt - came LAST and second-last. The trend is real in the sample and is not projectable. A hiring plan built on extrapolating it would be built on noise.

12 observations against a 12-week horizon. The 4-week view (394-439) is the defensible one; the 12-week flow interval spans a factor of 3.6 and I say so.

TASK 8

Similarity cannot be measured from titles

	M1 extended taxonomy	M2 pre-extension	Shift
NVIDIA - Microsoft	0.216 (closest)	0.093	+0.123
NVIDIA - Meta	0.147	0.195 (closest)	-0.048
Google - Microsoft, full text	0.57	stable	-

- **The ranking flips.** NVIDIA's closest peer is Microsoft on one matrix and Meta on the other. Changing the taxonomy moves a pair by more than most of the scores.
- Title vectors carry ~1 skill each - too sparse for cosine similarity to mean anything.
- The full-text matrix works (0.38-0.57) and is stable. **NVIDIA cannot appear in it.**
- **Recommendation: do not publish a four-company similarity matrix yet.**

TASK 9

What a decision-maker should do with this

If you are	Then
Competing for silicon talent	Expect ~416 standing NVIDIA roles turning over in ~2 weeks. Analog, mixed-signal and signal-integrity roles are ~90% senior - the contested segment is experienced hires
Watching NVIDIA's strategy	Track Solutions Architecture and Developer Relations. The shift suggests commercial deployment of AI infrastructure is scaling faster than research
Planning training	Inference and serving, compilers, agentic AI rising fastest - the deployment layer. AI Safety went 0 → 7, the cleanest signal in the set
Benchmarking salary	Not possible. No salary field for NVIDIA in this source

HONESTY

What this data cannot tell you

- **Closed does not mean filled.** Nothing separates a filled role from a cancelled or relisted one. Fast turnover is not evidence of fast hiring.
- **European bias.** The source curates EU-office employers. This is NVIDIA's European hiring.
- **Titles, not descriptions.** A skill absent here means the title did not say it, not that NVIDIA does not need it.
- **Four months, 12 observations.** No seasonality; the 12-week forecast is too wide to plan against.
- **Third-party derived.** NVIDIA's own terms ruled out first-party collection.

Why this slide exists

Every one of these is in the committed reports, not just the deck. A mentor should be able to check any number in this presentation against a file in the repository - and each report names the file.

NEXT

What the team should do, and questions

Priority	Action	Unblocks
1	Every member extends the taxonomy for their own company, then all re-run Task 4	Tasks 6 and 8 - currently not publishable
2	Decide titles or descriptions once, for everyone	Like-for-like comparison across all four
3	Google and Microsoft add posting dates; align Meta's 2024 window	Any cross-company time series or forecast
4	Re-run NVIDIA collection at --every 2 (60 observations instead of 12)	Task 7 - the single biggest improvement available

Expected questions: why no description text · why the taxonomy was extended unilaterally · why SES over ARIMA or Prophet · whether the flat stock is real or an EU artefact · what "closed" actually means.